

Supplemental Material

Finite-width origin of parametric self-limitation in driven nonlinear oscillators

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This document describes the numerical methods, the pre-specified acceptance criteria, the convergence tests, and the negative results referenced in the main text. Two accompanying code files reproduce all results: *SM_grid_engine.py* (the validated grid split-operator engine, v1.0) and *SM_reproduce_figures.py* (which reproduces Figs. 1–4 by calling the engine). Environment: Python 3.11, numpy ≥ 1.24 , matplotlib ≥ 3.7 . Both files are openly available in the repository cited in the main text.

S1. Grid split-operator method

We solve the single-particle Schrödinger equation for a driven anharmonic oscillator on a uniform position grid, using a symmetric (Strang) split-operator propagator evaluated with the fast Fourier transform. The effective- $\hbar$ form of the equation of motion is

$$i \hbar_{\text{eff}} \partial_t \psi = [T + V(x,t)] \psi, \quad T = -(\hbar_{\text{eff}}^2/2) \partial_{xx}, \quad V(x,t) = \frac{1}{2} \Omega^2(t) x^2 + V_{\text{nl}}(x),$$

with the parametric drive $\Omega^2(t) = \Omega_0^2(1 + A \cos \omega_p t)$ at the principal resonance $\omega_p = 2\Omega_0$. The nonlinear part V_{nl} is zero (harmonic reference), $+(\beta/4)x^4$ (confining Kerr), $+(\beta/4)x^4 + c_6x^6$ (steeper confining), or the cosine well $-\Omega^2(t)\cos x$ (softening/escape). The anharmonicity is normalized as $|U| = |\text{quartic coefficient}|/\Omega_0^2 = \beta/4$. One Strang step advances the state by a half potential phase (diagonal in position space), a full kinetic phase applied in Fourier space, and a second half potential phase; the local error is $O(\Delta t^3)$. The propagator is unitary and unconditionally stable, and — because the potential is applied as a diagonal multiplication in position space — it is free of the basis-edge instabilities that arise when the same strongly anharmonic driven Hamiltonian is represented in a truncated Fock basis. The instantaneous ground state is obtained by imaginary-time propagation of the same split-operator with $\Delta t \rightarrow -i\Delta t$. Default parameters: $L = 24$, $M = 4096$, $\Delta t = 1.5\text{--}2 \times 10^{-3}$, $T = 30$ (units of Ω_0^{-1}); for the softening/escape platform $L = 48$ and $M = 8192$ are required (Sec. S6).

S2. Parametric-gain observable

The instability rate is the parametric gain, defined as the secular exponential growth rate of the largest eigenvalue $\sigma_{\text{max}}(t)$ of the single-mode covariance matrix $C(t) = [[\text{Var}_x, \text{Cov}_{xp}], [\text{Cov}_{xp}, \text{Var}_p]]$, with all moments computed from the evolving grid state. This is the quantum counterpart of the largest classical tangent-space stretching, and it reduces to it in the classical limit (Sec. S4).

Canonical variables. The canonical momentum of the equation of motion of Sec. S1 is $p = -i\hbar_{\text{eff}}\partial_x$, so the momentum moments carry explicit factors of $\hbar_{\text{eff}}$: $\langle p \rangle = \hbar_{\text{eff}}\langle k \rangle$ and $\langle p^2 \rangle = \hbar_{\text{eff}}^2\langle k^2 \rangle$ in Fourier space. Omitting them replaces Var_p by the wavenumber variance, larger by $1/\hbar_{\text{eff}}^2$, and places the quantum and classical branches on different metrics. The factor equals unity at $\hbar_{\text{eff}} = 1$, so a validation performed only at that width cannot detect this class of error; the initial condition $\text{Var}_x = \text{Var}_p = \hbar_{\text{eff}}/2$ is therefore verified at every width used (Table S1).

We report the normalized gain $F = \lambda_{\text{param}}/\langle h \rangle$, where $h(t) = \eta(t)\Omega(t)$, $\eta = \omega^{-2}|d\Omega/dt|$, and $\langle h \rangle$ is the time average of h over the fit window actually used by the rate extractor, so that numerator and denominator refer to the same interval in every branch. The rate is extracted by a period-averaged linear fit of $\ln \sigma_{\text{max}}$ (Sec. S5).

Analytic normalization. For $\Omega^2(t) = \Omega_0^2(1 + A \cos 2\Omega_0 t)$ one has $\Omega d\Omega/dt = -A\Omega_0^3 \sin 2\Omega_0 t$ exactly, so $h(t) = A\Omega_0|\sin 2\Omega_0 t|$ with no corrections in A , and $\langle h \rangle = (2/\pi)A\Omega_0$. With the Mathieu rate $\lambda = A\Omega_0/4$ the harmonic gain ratio is $F_{\text{harm}} = \pi/8 = 0.392699$, independent of the drive amplitude and of $\hbar_{\text{eff}}$. Had $\langle h \rangle$ been defined as the root mean square of h over the same window, the identical construction would give

$1/\sqrt{8} = 0.353553$, lower by the factor $\pi\sqrt{2}/4 = 1.111$; the time-average convention is used throughout, for every branch, on the same window.

Validation. For the harmonic oscillator the covariance top-eigenvalue grows at the analytic Mathieu rate: the extracted λ_{param} equals $A/4$ to better than 1% with $R^2 = 1.00$ and window spread $< 0.5\%$, and the harmonic ground-state energy equals $\hbar_{\text{eff}}/2$ to four digits (Fig. 1). The corresponding normalized gain is $F = 0.390$ against the analytic $\pi/8 = 0.393$, at every width tested.

S3. Why the covariance observable?

We initially investigated several candidate observables for the instability rate, including out-of-time-ordered correlators (OTOCs) and direct occupation-number growth. For the present quasi-integrable single-mode systems, these failed at least one of the pre-specified acceptance criteria — log-linearity, window stability, or convergence (Sec. S8). The covariance-top-eigenvalue observable was adopted because it (1) reproduces the exact Mathieu rate in the harmonic limit (Sec. S2); (2) continuously approaches the classical tangent-stretching rate as $\hbar_{\text{eff}} \rightarrow 0$ (Sec. S4, Fig. 3); and (3) satisfies all pre-specified validation criteria (Sec. S5). The observable was therefore fixed by the validation procedure rather than chosen to obtain a particular answer.

S4. Effective $\hbar$ and the control ratio χ

Quantumness is tuned by $\hbar_{\text{eff}}$ in the equation of motion of Sec. S1: the vacuum width scales as $\sqrt{\hbar_{\text{eff}}}$, while the physical parameters (Ω_0 , $|U|$, and hence $\xi = \eta/|U|$) are held fixed. The classical limit is $\hbar_{\text{eff}} \rightarrow 0$. The reference classical rate is the finite-time tangent-stretching rate of the corresponding classical driven oscillator (function *classical_tangent* in the engine).

The tangent estimator requires a fit window that excludes the initial interval during which the tangent vector aligns with the growing direction. A fit that includes it underestimates the rate by an amount set by the window length alone: for the harmonic platform at $A = 0.20$, a window $\Omega_0 t \in [2, 30]$ returns $F = 0.335$, whereas $\Omega_0 t \in [40, 200]$ returns $F = 0.3925$, against the analytic $\pi/8 = 0.3927$. The two classical estimators — ensemble covariance and single-trajectory tangent — must coincide in the harmonic limit, and with a converged window they agree to 0.3% (0.3936 and 0.3925). Reported values use $T_{\text{win}} \geq 100$ with the fit starting at $\geq 0.2 T_{\text{win}}$.

We characterize the crossover by the dimensionless ratio $\chi \equiv \hbar_{\text{eff}} |U|$ ($\propto$ vacuum-width² / anharmonic-scale²). The numerical results are consistent with χ acting as the relevant control parameter within the explored parameter range. This was checked along two axes: varying $\hbar_{\text{eff}}$ at fixed $|U|$ (Fig. 3), and varying $|U|$ at fixed $\hbar_{\text{eff}}$. The second axis is drive-amplitude-limited at small χ and is therefore reported qualitatively; we do not claim more than consistency with χ as the controlling ratio.

S5. Pre-specified acceptance criteria

All acceptance criteria were fixed before the production runs, to avoid fitting an ambiguous slope. The raw $\sigma_{\text{max}}(t)$ oscillates at the drive period; we first remove this by a moving average over one period ($= \pi$ in these units) and fit $\ln \sigma_{\text{max}}$ on the envelope. A gain value is accepted only if all of the following hold:

(W1) Log-linearity: the fit has $R^2 \geq 0.98$.

(W2) Window independence: the slope is stable to $\leq 10\%$ across sub-windows.

(W3) Separation from saturation: the window ends before σ_{max} first reaches 60% of its run maximum. The criterion refers to the first crossing of that threshold: σ_{max} is not monotonic once the gain self-limits, so points on the descending branch also lie below the threshold, and taking the last such point would reopen the window past the maximum.

(G) Grid/step convergence: stable to $\leq 10\%$ under variation of L , M , Δt (Sec. S6).

Values failing these criteria are reported as such (open markers in the figures) and are not used to support any quantitative claim. Parameter values at which no window satisfies (W1)–(W3) are marked separately and no rate is quoted; at $\hbar_{\text{eff}} = 1$ the growth over the integration window is too small for a rate to be accepted.

S6. Convergence and validation checks

Grid and step convergence were checked at the crossover point $\zeta = 3$ over the ranges below. The confining parametric gain is stable to $< 5\%$ across all combinations; the harmonic control (Sec. S2) is exact ($R^2 = 1.00$) throughout.

Two further checks are reported because the harmonic validation alone does not cover them. The initial-state check verifies the canonical normalization of the covariance at every width. The box-size check is required for the softening/escape platform, whose over-barrier population wraps around the periodic grid and re-enters: at $L = 24$ the extracted rate is not converged ($F = 0.293$ at $L = 24$ against 0.151 at $L = 48$, with the sub-window spread falling from 0.81 to 0.04), while it is already converged in M .

TABLE S1. Convergence and validation checks.

Check	Values tested / expected	Outcome
Grid half-width L	20, 24, 28, 32 (confining)	$< 5\%$
Grid half-width L (escape)	24, 36, 48	$L \geq 48$ required; see text
Grid points M	1536, 2048, 3072, 4096 (8192 for escape)	$< 5\%$
Time step Δt	1, 1.5, 2, 3×10^{-3}	$< 1\%$
Harmonic ground-state energy	$\hbar_{\text{eff}}/2$	4 significant digits
Parametric gain, harmonic	Mathieu $A/4$; $F = \pi/8$	$< 1\%$; $F = 0.390$
Initial covariance, every $\hbar_{\text{eff}}$	$\text{Var } x = \text{Var } p = \hbar_{\text{eff}}/2$	holds at every width
Two classical estimators	must coincide in harmonic limit	0.3936 vs 0.3925

S7. Classical-ensemble (truncated-Wigner) benchmark

To separate the classical finite-initial-width contribution to the self-limitation from any genuine quantum correction, we evolve a classical (truncated-Wigner) ensemble: N classical trajectories whose initial Gaussian phase-space distribution is matched to the quantum vacuum ($\text{Var}_x = \text{Var}_p = \hbar_{\text{eff}}/2$), integrated by RK4 under the same drive, from which the same covariance-top-eigenvalue observable (Sec. S2), the same acceptance criteria (Sec. S5) and the same fit window are applied. For quadratic Hamiltonians the Wigner function evolves exactly under the classical Liouville equation; the same initial Wigner distribution is propagated classically in the weakly anharmonic regime as well, as the matched-width classical benchmark.

Control. For the harmonic oscillator the ensemble reproduces the analytic Mathieu rate $A/4$ at every width, as does the quantum branch ($F = 0.390$ against the analytic $\pi/8 = 0.393$). This control is repeated at each $\hbar_{\text{eff}}$ rather than at $\hbar_{\text{eff}} = 1$ alone, since it is the only one sensitive to the momentum normalization of Sec. S2.

Convergence. The confining-Kerr gain is stable in trajectory number ($< 1\%$ for $N \geq 2000$) and across random seeds (standard deviation ≈ 0.004).

Result. With the initial width matched and both branches evaluated in the same canonical variables, on the same observable and the same fit window, the classical ensemble reproduces the quantum parametric gain across the whole crossover:

$\hbar_{\text{eff}}$	F quantum	F classical ensemble	$\Delta F = F_{\text{qm}} - F_{\text{cl}}$
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0.50	0.3384	0.3440	-0.0056
0.25	0.3359	0.3347	+0.0013
0.125	0.3590	0.3567	+0.0023
0.06	0.3725	0.3716	+0.0009

The two branches agree to within 0.2% at every accepted point, against the harmonic reference $\pi/8 = 0.393$. The differences are at the level of the numerical uncertainty of the rate extraction, estimated from the grid, time-step and window convergence of Sec. S6 and from the trajectory-number and seed stability above. Within the validated observable and this numerical accuracy, the observed self-limitation is fully accounted for by classical finite-width dynamics, and no additional quantum correction beyond the initial width is resolved. The point at $\hbar_{\text{eff}} = 0.50$ lies slightly outside that band; at this width the growth departs from a clean exponential earlier, and the extraction is correspondingly less stable. The implementation is the function *classical_ensemble* in *SM_grid_engine.py*.

S8. Negative results

Two negative results were enforced by the criteria of Sec. S5 and are reported for completeness.

(a) Out-of-time-ordered correlator (OTOC). An OTOC estimate of the same instability rate, $C(t) = \|[x(t), p]|\psi_0\|^2$ evaluated by forward/backward evolution, does not yield an accepted rate for these driven single-mode systems: the confining case is grid-converged but fails log-linearity ($R^2 \approx 0.10$) and window independence ($\approx 13\%$), and the softening case fails Hilbert-space convergence. These single-mode driven oscillators are quasi-integrable, so the OTOC has no clean exponential regime; we therefore use the covariance-gain observable (Sec. S2, S3).

(b) Internal gain vs escape (softening well). At the pre-specified $|U| = 1/24$ the softening (cosine) well is too shallow to support a cleanly separable internal exponential amplification: internal gain and over-barrier leakage overlap. This is not a consequence of the parameter range explored but a structural property of the model. For $V(x) = \frac{1}{2}\Omega_0^2 x^2 - (\beta/4)x^4$ the barrier height is $V_b = \Omega_0^2/(16|U|)$, so the number of quanta below the barrier obeys $N_b = V_b/(\hbar_{\text{eff}}\Omega_0) = 1/(16\chi)$, i.e. $N_b \cdot \chi = 1/16$ identically. Once m and Ω_0 are fixed the potential retains a single independent shape parameter, so the well depth and the crossover parameter are not independent axes: the accessible parameter space is a curve, not a plane, and no scan over β and $\hbar_{\text{eff}}$ within this model can separate the two contributions. A controlled escape study therefore requires a second independent shape parameter. The obvious candidate fails, and for an analytic reason: a barrier-shaping sextic term leaves $|U| \equiv \beta/(4\Omega_0^2)$ unaltered, since γ enters only at order x^6 , but the barrier survives only while the two positive roots of $\gamma u^2 - \beta u + \Omega_0^2 = 0$ in $u = x^2$ remain distinct, that is for $\gamma < \beta^2/4\Omega_0^2$; at the bound they merge and the barrier disappears. The admissible range of γ is therefore closed, and scanning all of it changes N_b at fixed χ by a factor of only 1.3. What is required is a change of potential class. The two-harmonic well $V_0(x) = a[1 - \cos kx] + b[1 - \cos 2kx]$, with a and b solved from $k^2(a + 4b) = \Omega_0^2$ and $k^4(a + 16b)/24 = |U|\Omega_0^2$, leaves the period k free as an independent depth axis: $N_b \chi = 0.111, 0.083, 0.039$ and 0.017 at $k = 0.7, 1.0, 1.6$ and 2.5 , against the constant $1/16 = 0.0625$ of the quartic well, and V_b spans a factor of 6.5 at unchanged $|U|$. The potential is bounded, so no absorbing boundary is required. Finally, this constraint is specific to the statically defined softening well under frequency modulation; it does not apply to driven bistable or Kerr-parametric descriptions in which the pump reshapes the effective quasienergy landscape and the barrier is dynamical.

Code files

SM_grid_engine.py (v1.0) — grid construction, imaginary-time ground state, split-operator time evolution with $\hbar_{\text{eff}}$, covariance-top-eigenvalue observable in canonical variables, pre-specified secular-rate extractor returning the fit window, window-averaged $\langle h \rangle$, classical-tangent reference, and the classical truncated-Wigner ensemble benchmark. Runs a harmonic self-test — including both classical controls — when executed directly. Environment: Python 3.11, numpy ≥ 1.24 .

SM_reproduce_figures.py — reproduces Figs. 1–4 by calling the engine. Environment adds matplotlib $\geq$ 3.7.